ETALE LIFTS OF APPROXIMATIONS ACROSS NILPOTENT THICKENINGS WITH PRESCRIBED SPECIAL FIBRE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a distinguished-square recognition on the special fibre, a deformation of the etale leg by topological invariance, a pushout construction pinned by base-change uniqueness, and an affineness conclusion, all inside the delivered span

Editorial note. The label lemma-lift-approximation names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The proof closes with the author's sentence 'It is clear that B fits into the desired diagram and satisfies all the properties required of it.'; the properties were established in the course of the construction, and this closing verification is retained verbatim.

Allowed standard prerequisites (editorial).

- Elementary distinguished squares of algebraic spaces and their base change stability
- Topological invariance of the etale site under finite order thickenings, with separatedness deformed along thickenings
- The pushout construction of elementary distinguished squares in the pushouts theory, with the identification of the pushout after base change, and affineness transfer from Limits of Spaces

Source reference. Original theorem 039R: sources/stacks-chow/tags.txt, source lines 4282–4282. Prerequisite pointer invoked at proof line(s) 2219; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 06AG: sources/stacks-chow/tags.txt, source lines 8075–8075. Prerequisite pointer invoked at proof line(s) 2222; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 07VT: sources/stacks-chow/tags.txt, source lines 10018–10018. Prerequisite pointer invoked at proof line(s) 2249; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 08GM: sources/stacks-chow/tags.txt, source lines 10750–10750. Prerequisite pointer invoked at proof line(s) 2210; it names a

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standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08GN: sources/stacks-chow/tags.txt, source lines 10751–10751. Prerequisite pointer invoked at proof line(s) 2235; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DVJ: sources/stacks-chow/tags.txt, source lines 17364–17364. Prerequisite pointer invoked at proof line(s) 2224; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DVI: sources/stacks-chow/tags.txt, source lines 17363–17363. Prerequisite pointer invoked at proof line(s) 2247; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let A be a Noetherian ring and $I \subset A$ an ideal. Let $J \subset A$ be a nilpotent ideal. Consider a commutative diagram*

$$\begin{array}{ccccc} C & \longrightarrow & C_0 & = & C/JC \\ \uparrow & & \uparrow & & \\ & & B_0 & & \\ & & \uparrow & & \\ A & \longrightarrow & A_0 & = & A/J \end{array}$$

whose vertical arrows are of finite type such that

1. $\mathrm{Spec}(C) \to \mathrm{Spec}(A)$ *is étale over* $\mathrm{Spec}(A) \setminus V(I)$,
2. $\mathrm{Spec}(B_0) \to \mathrm{Spec}(A_0)$ *is étale over* $\mathrm{Spec}(A_0) \setminus V(IA_0)$, *and*
3. $B_0 \to C_0$ *is étale and induces an isomorphism* $B_0/IB_0 = C_0/IC_0$.

Then we can fill in the diagram above to a commutative diagram

$$\begin{array}{ccc} C & \longrightarrow & C/JC \\ \uparrow & & \uparrow \\ B & \longrightarrow & B_0 \\ \uparrow & & \uparrow \\ A & \longrightarrow & A/J \end{array}$$

with $A \to B$ of finite type, $B/JB = B_0$, $B \to C$ étale, and $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ *étale over* $\mathrm{Spec}(A) \setminus V(I)$.

Proof. Set $X = \mathrm{Spec}(A)$, $X_0 = \mathrm{Spec}(A_0)$, $Y_0 = \mathrm{Spec}(B_0)$, $Z = \mathrm{Spec}(C)$, $Z_0 = \mathrm{Spec}(C_0)$. Furthermore, denote $U \subset X$, $U_0 \subset X_0$, $V_0 \subset Y_0$, $W \subset Z$, $W_0 \subset Z_0$ the complement of the vanishing set of I. Here is a picture to help visualize the

situation:

$$\begin{array}{ccc} Z & \leftarrow & Z_0 \\ \downarrow & & \downarrow \\ & & Y_0 \\ & & \downarrow \\ X & \leftarrow & X_0 \end{array} \qquad \begin{array}{ccc} W & \leftarrow & W_0 \\ \downarrow & & \downarrow \\ & & V_0 \\ & & \downarrow \\ U & \leftarrow & U_0 \end{array}$$

The conditions in the lemma guarantee that

$$\begin{array}{ccc} W_0 & \longrightarrow & Z_0 \\ \downarrow & & \downarrow \\ V_0 & \longrightarrow & Y_0 \end{array}$$

is an elementary distinguished square, see Derived Categories of Spaces, Definition 08GM. In addition we know that $W_0 \to U_0$ and $V_0 \to U_0$ are étale. The morphism $X_0 \subset X$ is a finite order thickening as J is assumed nilpotent. By the topological invariance of the étale site we can find a unique étale morphism $V \to X$ of schemes with $V_0 = V \times_X X_0$ and we can lift the given morphism $W_0 \to V_0$ to a unique morphism $W \to V$ over X. See Étale Morphisms, Theorem 039R. Since $W_0 \to V_0$ is separated, the morphism $W \to V$ is separated too, see for example More on Morphisms, Lemma 06AG. By Pushouts of Spaces, Lemma 0DVJ we can construct an elementary distinguished square

$$\begin{array}{ccc} W & \longrightarrow & Z \\ \downarrow & & \downarrow \\ V & \longrightarrow & Y \end{array}$$

in the category of algebraic spaces over X. Since the base change of an elementary distinguished square is an elementary distinguished square (Derived Categories of Spaces, Lemma 08GN) we see that

$$\begin{array}{ccc} W_0 & \longrightarrow & Z_0 \\ \downarrow & & \downarrow \\ V_0 & \longrightarrow & Y \times_X X_0 \end{array}$$

is an elementary distinguished square. It follows that there is a unique isomorphism $Y \times_X X_0 = Y_0$ compatible with the two squares involving these spaces because elementary distinguished squares are pushouts (Pushouts of Spaces, Lemma 0DVI). It follows that Y is affine by Limits of Spaces, Proposition 07VT. Write $Y = \mathrm{Spec}(B)$. It is clear that B fits into the desired diagram and satisfies all the properties required of it. $\square$